

Formative Assignment 0 - Solutions

The Untyped λ -Calculus

1. (a) We underline the β -redexes reduced in each step:

$$\begin{aligned}
 & \text{add } \underline{2} \ \underline{3} \\
 & \mapsto_{\beta} (\lambda b. \lambda f. \lambda x. \underline{2} \ f \ (b \ f \ x)) \ \underline{3} \\
 & \mapsto_{\beta} \lambda f. \lambda x. \underline{2} \ f \ (\underline{3} \ f \ x) \\
 & \mapsto_{\beta} \lambda f. \lambda x. (\lambda x. f(fx)) \ (\underline{3} \ f \ x) \\
 & \mapsto_{\beta} \lambda f. \lambda x. f(f(\underline{3} \ f \ x)) \\
 & \mapsto_{\beta} \lambda f. \lambda x. f(f((\lambda x. f(f(fx))) \ x)) \\
 & \mapsto_{\beta} \lambda f. \lambda x. f(f(f(f(fx))))
 \end{aligned}$$

- (b) Let $\text{mul} = \lambda a. \lambda b. \lambda f. \lambda x. a \ (b \ f) \ x$. We underline the β -redexes reduced in each step:

$$\begin{aligned}
 & \text{mul } \underline{2} \ \underline{3} \\
 & \mapsto_{\beta} (\lambda b. \lambda f. \lambda x. \underline{2} \ (b \ f) \ x) \ \underline{3} \\
 & \mapsto_{\beta} \lambda f. \lambda x. \underline{2} \ (\underline{3} \ f) \ x \\
 & \mapsto_{\beta} \lambda f. \lambda x. (\lambda x. (\underline{3} \ f) ((\underline{3} \ f) x)) \ x \\
 & \mapsto_{\beta} \lambda f. \lambda x. (\underline{3} \ f) ((\underline{3} \ f) x) \\
 & \mapsto_{\beta} \lambda f. \lambda x. (\lambda x. f(f(fx))) ((\underline{3} \ f) x) \\
 & \mapsto_{\beta} \lambda f. \lambda x. f(f(f((\underline{3} \ f) x))) \\
 & \mapsto_{\beta} \lambda f. \lambda x. f(f(f((\lambda x. f(f(fx))) x))) \\
 & \mapsto_{\beta} \lambda f. \lambda x. f(f(f(f(f(fx)))))
 \end{aligned}$$

2. The syntax can be extended as follows:

$$M ::= x \mid \lambda x. M \mid M \ M \mid \text{zero} \mid \text{succ}(M) \mid \text{pred}(M) \mid \text{iszero}(M, M, M)$$

The β -reduction relation can be extended with the following axioms:

$$\begin{aligned}
 & \text{pred}(\text{zero}) \mapsto_{\beta} \text{zero} \\
 & \text{pred}(\text{succ}(M)) \mapsto_{\beta} M \\
 & \text{iszero}(\text{zero}, N, P) \mapsto_{\beta} N \\
 & \text{iszero}(\text{succ}(M), N, P) \mapsto_{\beta} P
 \end{aligned}$$

and rules:

$$\begin{aligned}
 & \text{if } M \mapsto_{\beta} N \text{ then } \text{pred}(M) \mapsto_{\beta} \text{pred}(N) \\
 & \text{if } M \mapsto_{\beta} N \text{ then } \text{iszero}(M, P, Q) \mapsto_{\beta} \text{iszero}(N, P, Q)
 \end{aligned}$$

3. Let

$$\begin{aligned}
F &= \lambda f. \lambda a. \lambda b. \text{iszero}(a, b, f \text{ pred}(a) \text{ succ}(b)) \\
\text{add}' &= Y F \\
G &= (\lambda x. F(xx))(\lambda x. F(xx))
\end{aligned}$$

We include the detail calculation for reference:

$$\begin{aligned}
\text{add}' \ \bar{2} \ \bar{3} &\mapsto_{\beta} \underline{G \ \bar{2} \ \bar{3}} \\
&\mapsto_{\beta} \underline{F \ G \ \bar{2} \ \bar{3}} \\
&\mapsto_{\beta} \underline{(\lambda a. \lambda b. \text{iszero}(a, b, G \text{ pred}(a) \text{ succ}(b))) \ \bar{2} \ \bar{3}} \\
&\mapsto_{\beta} \underline{(\lambda b. \text{iszero}(\bar{2}, b, G \text{ pred}(\bar{2}) \text{ succ}(b))) \ \bar{3}} \\
&\mapsto_{\beta} \underline{\text{iszero}(\bar{2}, \bar{3}, G \text{ pred}(\bar{2}) \ \bar{4})} \\
&\mapsto_{\beta} \underline{G \text{ pred}(\bar{2}) \ \bar{4}} \\
&\mapsto_{\beta} \underline{F \ G \text{ pred}(\bar{2}) \ \bar{4}} \\
&\mapsto_{\beta} \underline{(\lambda a. \lambda b. \text{iszero}(a, b, G \text{ pred}(a) \text{ succ}(b))) \text{ pred}(\bar{2}) \ \bar{4}} \\
&\mapsto_{\beta} \underline{(\lambda b. \text{iszero}(\text{pred}(\bar{2}), b, G \text{ pred}(\text{pred}(\bar{2})) \text{ succ}(b))) \ \bar{4}} \\
&\mapsto_{\beta} \underline{\text{iszero}(\text{pred}(\bar{2}), \bar{4}, G \text{ pred}(\text{pred}(\bar{2})) \ \bar{5})} \\
&\mapsto_{\beta} \underline{\text{iszero}(\bar{1}, \bar{4}, G \text{ pred}(\text{pred}(\bar{2})) \ \bar{5})} \\
&\mapsto_{\beta} \underline{G \text{ pred}(\text{pred}(\bar{2})) \ \bar{5}} \\
&\mapsto_{\beta} \underline{F \ G \text{ pred}(\text{pred}(\bar{2})) \ \bar{5}} \\
&\mapsto_{\beta} \underline{(\lambda a. \lambda b. \text{iszero}(a, b, G \text{ pred}(a) \text{ succ}(b))) \text{ pred}(\text{pred}(\bar{2})) \ \bar{5}} \\
&\mapsto_{\beta} \underline{(\lambda b. \text{iszero}(\text{pred}(\text{pred}(\bar{2})), b, G \text{ pred}(\text{pred}(\text{pred}(\bar{2}))) \text{ succ}(b))) \ \bar{5}} \\
&\mapsto_{\beta} \underline{\text{iszero}(\text{pred}(\text{pred}(\bar{2})), \bar{5}, G \text{ pred}(\text{pred}(\text{pred}(\bar{2}))) \ \bar{6})} \\
&\mapsto_{\beta} \underline{\text{iszero}(\text{pred}(\bar{1}), \bar{5}, G \text{ pred}(\text{pred}(\text{pred}(\bar{2}))) \ \bar{6})} \\
&\mapsto_{\beta} \underline{\text{iszero}(\text{zero}, \bar{5}, G \text{ pred}(\text{pred}(\text{pred}(\bar{2}))) \ \bar{6})} \\
&\mapsto_{\beta} \bar{5}
\end{aligned}$$